| No. | Status | Date |
|---|---|---|
| LNDE-RPT-96-014-DS-4-LA-102 | Issued | 1996-03-08 |

# UT Thickness Data Sheet - 4-LA-102

## Scope and Responsibility

This data sheet provides the ultrasonic thickness measurements for line 4-LA-102, located within the ATU-100. The scope of work is limited to the baseline thickness recording of this specific lean amine supply line to establish initial wall thickness data prior to unit service.

Grace Okafor, acting as the UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., is responsible for the execution of the measurements and the recording of the data presented herein.

## Technical Specifications

Line 4-LA-102 is constructed from A106 Gr.B with an outer diameter of 114.3 mm and a nominal thickness of 6.02 mm. The minimum allowable thickness (t-min) is 3.18 mm.

The design parameters for this line are established at a maximum allowable pressure of 2413 kPa(g) and a maximum allowable temperature of 93 °C.

## Inspection Methodology

Thickness measurements were performed in accordance with QSCP-NDT-001 Rev 0. The inspection was conducted as part of the pre-startup baseline survey for the ATU-100.

The ultrasonic testing was carried out citing the requirements of ASTM E797 to ensure measurement accuracy and consistency across the identified monitoring points.

## Data Summary and Reference

The measurements recorded on this data sheet are extracted from LNDE-RPT-96-014, which details the baseline ultrasonic thickness survey for 118 CMLs. As the unit is not yet in service, these values represent the initial state of the asset.

All readings are compared against the t-min of 3.18 mm. Any reading falling below this threshold is subject to the site's non-conformance reporting process.

## Thickness readings, measured 1996-03-08

**Line 6-SG-101 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-101-01 | 7.11 | 3.56 | — | — | 7.12 | — | — | — | 2001-03-08 | |
| 6-SG-101-02 | 7.11 | 3.56 | — | — | 7.05 | — | — | — | 2001-03-08 | |
| 6-SG-101-03 | 7.11 | 3.56 | — | — | 7.08 | — | — | — | 2001-03-08 | |
| 6-SG-101-04 | 7.11 | 3.56 | — | — | 7.09 | — | — | — | 2001-03-08 | |
| 6-SG-101-05 | 7.11 | 3.56 | — | — | 7.14 | — | — | — | 2001-03-08 | |
| 6-SG-101-06 | 7.11 | 3.56 | — | — | 7.07 | — | — | — | 2001-03-08 | |

**Line 6-SG-102 | Sour gas inlet | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-SG-102-01 | 7.11 | 3.56 | — | — | 7.14 | — | — | — | 2001-03-08 | |
| 6-SG-102-02 | 7.11 | 3.56 | — | — | 7.11 | — | — | — | 2001-03-08 | |
| 6-SG-102-03 | 7.11 | 3.56 | — | — | 7.04 | — | — | — | 2001-03-08 | |
| 6-SG-102-04 | 7.11 | 3.56 | — | — | 6.99 | — | — | — | 2001-03-08 | |
| 6-SG-102-05 | 7.11 | 3.56 | — | — | 7.13 | — | — | — | 2001-03-08 | |

**Line 8-TG-101 | Treated gas | A106 Gr.B | t_nom 8.18 mm | t-min 4.06 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 8-TG-101-01 | 8.18 | 4.06 | — | — | 8.20 | — | — | — | 2006-03-08 | |
| 8-TG-101-02 | 8.18 | 4.06 | — | — | 8.20 | — | — | — | 2006-03-08 | |
| 8-TG-101-03 | 8.18 | 4.06 | — | — | 8.23 | — | — | — | 2006-03-08 | |
| 8-TG-101-04 | 8.18 | 4.06 | — | — | 8.10 | — | — | — | 2006-03-08 | |
| 8-TG-101-05 | 8.18 | 4.06 | — | — | 8.22 | — | — | — | 2006-03-08 | |

**Line 4-LA-101 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-101-01 | 6.02 | 3.18 | — | — | 5.99 | — | — | — | 2006-03-08 | |
| 4-LA-101-02 | 6.02 | 3.18 | — | — | 6.01 | — | — | — | 2006-03-08 | |
| 4-LA-101-03 | 6.02 | 3.18 | — | — | 5.96 | — | — | — | 2006-03-08 | |
| 4-LA-101-04 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-101-05 | 6.02 | 3.18 | — | — | 6.01 | — | — | — | 2006-03-08 | |
| 4-LA-101-06 | 6.02 | 3.18 | — | — | 6.06 | — | — | — | 2006-03-08 | |
| 4-LA-101-07 | 6.02 | 3.18 | — | — | 5.99 | — | — | — | 2006-03-08 | |

**Line 4-LA-102 | Lean amine supply | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-102-01 | 6.02 | 3.18 | — | — | 6.02 | — | — | — | 2006-03-08 | |
| 4-LA-102-02 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-102-03 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-102-04 | 6.02 | 3.18 | — | — | 5.99 | — | — | — | 2006-03-08 | |
| 4-LA-102-05 | 6.02 | 3.18 | — | — | 6.00 | — | — | — | 2006-03-08 | |
| 4-LA-102-06 | 6.02 | 3.18 | — | — | 6.00 | — | — | — | 2006-03-08 | |
| 4-LA-102-07 | 6.02 | 3.18 | — | — | 6.00 | — | — | — | 2006-03-08 | |

**Line 6-LA-103 | Lean amine header | A106 Gr.B | t_nom 7.11 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-LA-103-01 | 7.11 | 3.56 | — | — | 7.02 | — | — | — | 2006-03-08 | |
| 6-LA-103-02 | 7.11 | 3.56 | — | — | 7.19 | — | — | — | 2006-03-08 | |
| 6-LA-103-03 | 7.11 | 3.56 | — | — | 7.08 | — | — | — | 2006-03-08 | |
| 6-LA-103-04 | 7.11 | 3.56 | — | — | 7.17 | — | — | — | 2006-03-08 | |
| 6-LA-103-05 | 7.11 | 3.56 | — | — | 7.19 | — | — | — | 2006-03-08 | |
| 6-LA-103-06 | 7.11 | 3.56 | — | — | 7.16 | — | — | — | 2006-03-08 | |
| 6-LA-103-07 | 7.11 | 3.56 | — | — | 7.18 | — | — | — | 2006-03-08 | |
| 6-LA-103-08 | 7.11 | 3.56 | — | — | 7.04 | — | — | — | 2006-03-08 | |

**Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-101-01 | 8.56 | 3.81 | — | — | 8.52 | — | — | — | 2006-03-08 | |
| 4-RA-101-02 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-101-03 | 8.56 | 3.81 | — | — | 8.61 | — | — | — | 2006-03-08 | |
| 4-RA-101-04 | 8.56 | 3.81 | — | — | 8.63 | — | — | — | 2006-03-08 | |
| 4-RA-101-05 | 8.56 | 3.81 | — | — | 8.60 | — | — | — | 2006-03-08 | |
| 4-RA-101-07 | 8.56 | 3.81 | — | — | 8.60 | — | — | — | 2006-03-08 | |
| 4-RA-101-08 | 8.56 | 3.81 | — | — | 8.68 | — | — | — | 2006-03-08 | |
| 4-RA-101-09 | 8.56 | 3.81 | — | — | 8.57 | — | — | — | 2006-03-08 | |
| 4-RA-101-10 | 8.56 | 3.81 | — | — | 8.67 | — | — | — | 2006-03-08 | |
| 4-RA-101-11 | 8.56 | 3.81 | — | — | 8.68 | — | — | — | 2006-03-08 | |

**Line 4-RA-102 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-102-01 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-102-02 | 8.56 | 3.81 | — | — | 8.57 | — | — | — | 2006-03-08 | |
| 4-RA-102-03 | 8.56 | 3.81 | — | — | 8.56 | — | — | — | 2006-03-08 | |
| 4-RA-102-04 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-102-05 | 8.56 | 3.81 | — | — | 8.55 | — | — | — | 2006-03-08 | |
| 4-RA-102-06 | 8.56 | 3.81 | — | — | 8.51 | — | — | — | 2006-03-08 | |
| 4-RA-102-07 | 8.56 | 3.81 | — | — | 8.58 | — | — | — | 2006-03-08 | |
| 4-RA-102-08 | 8.56 | 3.81 | — | — | 8.49 | — | — | — | 2006-03-08 | |
| 4-RA-102-09 | 8.56 | 3.81 | — | — | 8.55 | — | — | — | 2006-03-08 | |

**Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 6-RA-103-01 | 10.97 | 4.57 | — | — | 11.00 | — | — | — | 2006-03-08 | |
| 6-RA-103-02 | 10.97 | 4.57 | — | — | 11.09 | — | — | — | 2006-03-08 | |
| 6-RA-103-03 | 10.97 | 4.57 | — | — | 10.99 | — | — | — | 2006-03-08 | |
| 6-RA-103-04 | 10.97 | 4.57 | — | — | 10.98 | — | — | — | 2006-03-08 | |
| 6-RA-103-05 | 10.97 | 4.57 | — | — | 10.95 | — | — | — | 2006-03-08 | |
| 6-RA-103-06 | 10.97 | 4.57 | — | — | 11.01 | — | — | — | 2006-03-08 | |
| 6-RA-103-07 | 10.97 | 4.57 | — | — | 10.99 | — | — | — | 2006-03-08 | |
| 6-RA-103-08 | 10.97 | 4.57 | — | — | 10.97 | — | — | — | 2006-03-08 | |
| 6-RA-103-09 | 10.97 | 4.57 | — | — | 11.01 | — | — | — | 2006-03-08 | |
| 6-RA-103-10 | 10.97 | 4.57 | — | — | 11.02 | — | — | — | 2006-03-08 | |

**Line 3-RV-101 | Regenerator overhead | A106 Gr.B | t_nom 7.62 mm | t-min 3.56 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RV-101-01 | 7.62 | 3.56 | — | — | 7.59 | — | — | — | 2001-03-08 | |
| 3-RV-101-02 | 7.62 | 3.56 | — | — | 7.57 | — | — | — | 2001-03-08 | |
| 3-RV-101-03 | 7.62 | 3.56 | — | — | 7.52 | — | — | — | 2001-03-08 | |
| 3-RV-101-04 | 7.62 | 3.56 | — | — | 7.59 | — | — | — | 2001-03-08 | |
| 3-RV-101-05 | 7.62 | 3.56 | — | — | 7.67 | — | — | — | 2001-03-08 | |
| 3-RV-101-06 | 7.62 | 3.56 | — | — | 7.59 | — | — | — | 2001-03-08 | |

**Line 2-RV-102 | Regenerator overhead | A106 Gr.B | t_nom 5.54 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RV-102-01 | 5.54 | 2.79 | — | — | 5.53 | — | — | — | 2001-03-08 | |
| 2-RV-102-02 | 5.54 | 2.79 | — | — | 5.59 | — | — | — | 2001-03-08 | |
| 2-RV-102-03 | 5.54 | 2.79 | — | — | 5.48 | — | — | — | 2001-03-08 | |
| 2-RV-102-04 | 5.54 | 2.79 | — | — | 5.58 | — | — | — | 2001-03-08 | |

**Line 3-RF-101 | Reflux | A312 TP316 | t_nom 5.49 mm | t-min 2.41 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-RF-101-01 | 5.49 | 2.41 | — | — | 5.48 | — | — | — | 2006-03-08 | |
| 3-RF-101-02 | 5.49 | 2.41 | — | — | 5.45 | — | — | — | 2006-03-08 | |
| 3-RF-101-03 | 5.49 | 2.41 | — | — | 5.45 | — | — | — | 2006-03-08 | |
| 3-RF-101-04 | 5.49 | 2.41 | — | — | 5.46 | — | — | — | 2006-03-08 | |
| 3-RF-101-05 | 5.49 | 2.41 | — | — | 5.45 | — | — | — | 2006-03-08 | |
| 3-RF-101-06 | 5.49 | 2.41 | — | — | 5.52 | — | — | — | 2006-03-08 | |
| 3-RF-101-07 | 5.49 | 2.41 | — | — | 5.42 | — | — | — | 2006-03-08 | |

**Line 2-RF-102 | Reflux | A312 TP316 | t_nom 3.91 mm | t-min 2.16 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-RF-102-01 | 3.91 | 2.16 | — | — | 3.92 | — | — | — | 2006-03-08 | |
| 2-RF-102-02 | 3.91 | 2.16 | — | — | 3.88 | — | — | — | 2006-03-08 | |
| 2-RF-102-03 | 3.91 | 2.16 | — | — | 3.93 | — | — | — | 2006-03-08 | |
| 2-RF-102-04 | 3.91 | 2.16 | — | — | 3.87 | — | — | — | 2006-03-08 | |
| 2-RF-102-07 | 3.91 | 2.16 | — | — | 3.96 | — | — | — | 2006-03-08 | |

**Line 4-LA-104 | L/R exchanger, lean side | A106 Gr.B | t_nom 6.02 mm | t-min 3.18 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-LA-104-01 | 6.02 | 3.18 | — | — | 6.07 | — | — | — | 2006-03-08 | |
| 4-LA-104-02 | 6.02 | 3.18 | — | — | 6.03 | — | — | — | 2006-03-08 | |
| 4-LA-104-03 | 6.02 | 3.18 | — | — | 5.96 | — | — | — | 2006-03-08 | |
| 4-LA-104-04 | 6.02 | 3.18 | — | — | 6.06 | — | — | — | 2006-03-08 | |
| 4-LA-104-05 | 6.02 | 3.18 | — | — | 6.04 | — | — | — | 2006-03-08 | |
| 4-LA-104-06 | 6.02 | 3.18 | — | — | 6.10 | — | — | — | 2006-03-08 | |
| 4-LA-104-07 | 6.02 | 3.18 | — | — | 6.01 | — | — | — | 2006-03-08 | |
| 4-LA-104-08 | 6.02 | 3.18 | — | — | 5.94 | — | — | — | 2006-03-08 | |

**Line 4-RA-104 | L/R exchanger, rich side | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 4-RA-104-01 | 8.56 | 3.81 | — | — | 8.51 | — | — | — | 2006-03-08 | |
| 4-RA-104-02 | 8.56 | 3.81 | — | — | 8.55 | — | — | — | 2006-03-08 | |
| 4-RA-104-03 | 8.56 | 3.81 | — | — | 8.54 | — | — | — | 2006-03-08 | |
| 4-RA-104-04 | 8.56 | 3.81 | — | — | 8.59 | — | — | — | 2006-03-08 | |
| 4-RA-104-05 | 8.56 | 3.81 | — | — | 8.60 | — | — | — | 2006-03-08 | |
| 4-RA-104-06 | 8.56 | 3.81 | — | — | 8.59 | — | — | — | 2006-03-08 | |
| 4-RA-104-07 | 8.56 | 3.81 | — | — | 8.59 | — | — | — | 2006-03-08 | |

**Line 2-ST-101 | Reboiler steam | A106 Gr.B | t_nom 3.91 mm | t-min 2.29 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 2-ST-101-01 | 3.91 | 2.29 | — | — | 3.93 | — | — | — | 2006-03-08 | |
| 2-ST-101-02 | 3.91 | 2.29 | — | — | 3.90 | — | — | — | 2006-03-08 | |
| 2-ST-101-03 | 3.91 | 2.29 | — | — | 3.94 | — | — | — | 2006-03-08 | |
| 2-ST-101-04 | 3.91 | 2.29 | — | — | 3.82 | — | — | — | 2006-03-08 | |
| 2-ST-101-05 | 3.91 | 2.29 | — | — | 3.91 | — | — | — | 2006-03-08 | |

**Line 3-CW-101 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-101-01 | 5.49 | 2.79 | — | — | 5.42 | — | — | — | 2006-03-08 | |
| 3-CW-101-02 | 5.49 | 2.79 | — | — | 5.41 | — | — | — | 2006-03-08 | |
| 3-CW-101-03 | 5.49 | 2.79 | — | — | 5.53 | — | — | — | 2006-03-08 | |
| 3-CW-101-04 | 5.49 | 2.79 | — | — | 5.46 | — | — | — | 2006-03-08 | |
| 3-CW-101-05 | 5.49 | 2.79 | — | — | 5.48 | — | — | — | 2006-03-08 | |

**Line 3-CW-102 | Amine cooler CW | A106 Gr.B | t_nom 5.49 mm | t-min 2.79 mm**

| CML | t_nom (mm) | t-min (mm) | Prev. date | Prev. (mm) | Current (mm) | CR LT (mm/y) | CR ST (mm/y) | RL (y) | Next due | Remark |
|---|---|---|---|---|---|---|---|---|---|---|
| 3-CW-102-01 | 5.49 | 2.79 | — | — | 5.50 | — | — | — | 2006-03-08 | |
| 3-CW-102-02 | 5.49 | 2.79 | — | — | 5.51 | — | — | — | 2006-03-08 | |
| 3-CW-102-03 | 5.49 | 2.79 | — | — | 5.53 | — | — | — | 2006-03-08 | |
| 3-CW-102-04 | 5.49 | 2.79 | — | — | 5.51 | — | — | — | 2006-03-08 | |

| Sign-off | Name | Function | Date | Signature |
|---|---|---|---|---|
| Prepared | Grace Okafor | UT technician, SNT-TC-1A UT Level II | 1996-03-08 | _____ |